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Drop Zone Criteria

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Purpose

- The purpose of this course is to prospective graduates on required criteria for a Landing Zone to be considered usable during Survey and Assault Zone Assessment (SAAZA).



Learning Objectives

- Identify the dimensions of the imaginary surface, to include slope ratio
- Identify the required criteria for an DZ



References

- UFC 3-260-1 Airfield and Heliport Planning and Design
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- Engineering Technical Letter 02-19 (12 Nov 02)



Outline

- Drop Zone (DZ)
- Review
- Summary



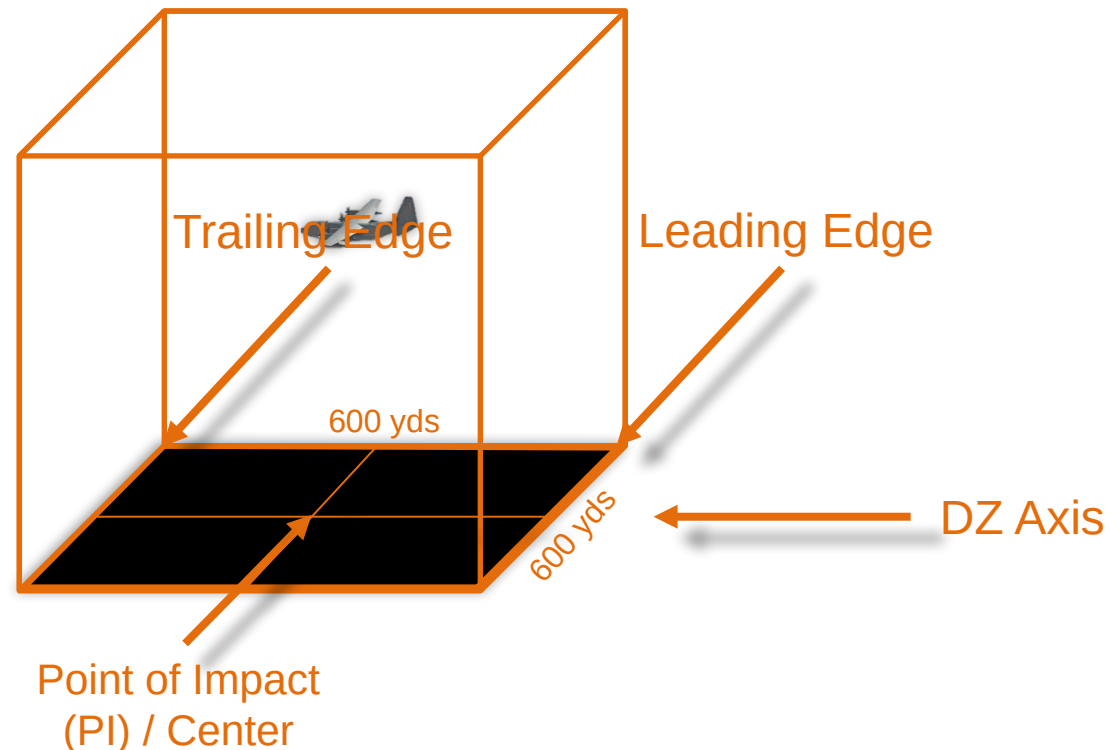
Outline

- Drop Zone (DZ)
- Summary



Drop Zone

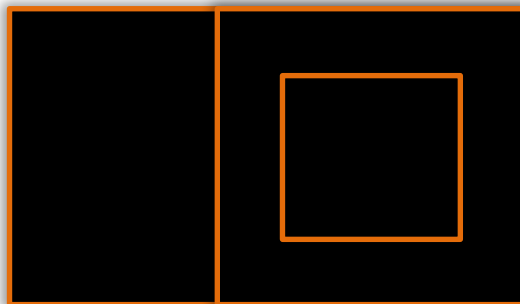
- Anatomy of a standard DZ
- DZ survey is three dimensional





Drop Zone

- Minimum criteria is based on mission type
 - Container Delivery System (CDS)
 - (400 yds width x 400 yds length x 600 ft altitude)
 - Personnel
 - (600 yds width x 600 yds length x 1000 ft altitude)
 - Heavy Equipment
 - (600 yds width x 1000 yds length x 1100 ft altitude)



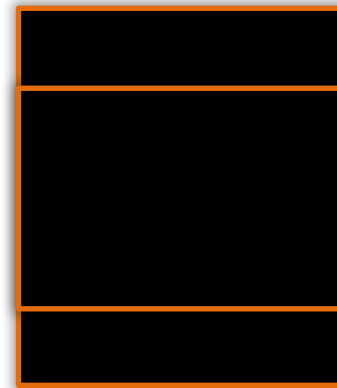


Drop Zone

- Criteria adjustments are based on mission specific information, and are cumulative
- Adjustments are made half to each side or end of a DZ, unless indicated

Adjustment requires
+200 yds width

100 yds to each side





Drop Zone

- Personnel (600 x 600 x 1000)
 - Add 75 yds to trailing edge for each jumper >1
 - For each 100 ft above 1000 ft, add 30 yds to length / width
 - Sunset to sunrise, add 100 yds to length
 - Day formations / night single, add 100 yds to width only
 - Night formation, add 200 yds to width only
 - GPS drops, add 400 yds to width, (exclude formation and night)



Drop Zones

- CDS (400 x 400 x 600)
 - Length based on number of containers
 - Day formations, add 100 yds to length / width
 - Sunset to sunrise, 100 yds for single / 200 yds for formation
 - For each 100 ft above 600 ft, add 40 yds
 - GPS, add 200 yds (exclude formation / night)



Drop Zone

- Heavy Equipment (600 x 1000 x 1100)
 - Add 400 yds to length for each platform >1
 - For each 100 ft above 1100 ft, add 30 yds to the width and length
 - Sunset to sunrise, add 100 yds to length
 - Day formation / night single, add 100 yds to width
 - Night formation, add 200 yds to width only
 - GPS drops, add 400 yds to width (exclude formation and night)



Point of Impact

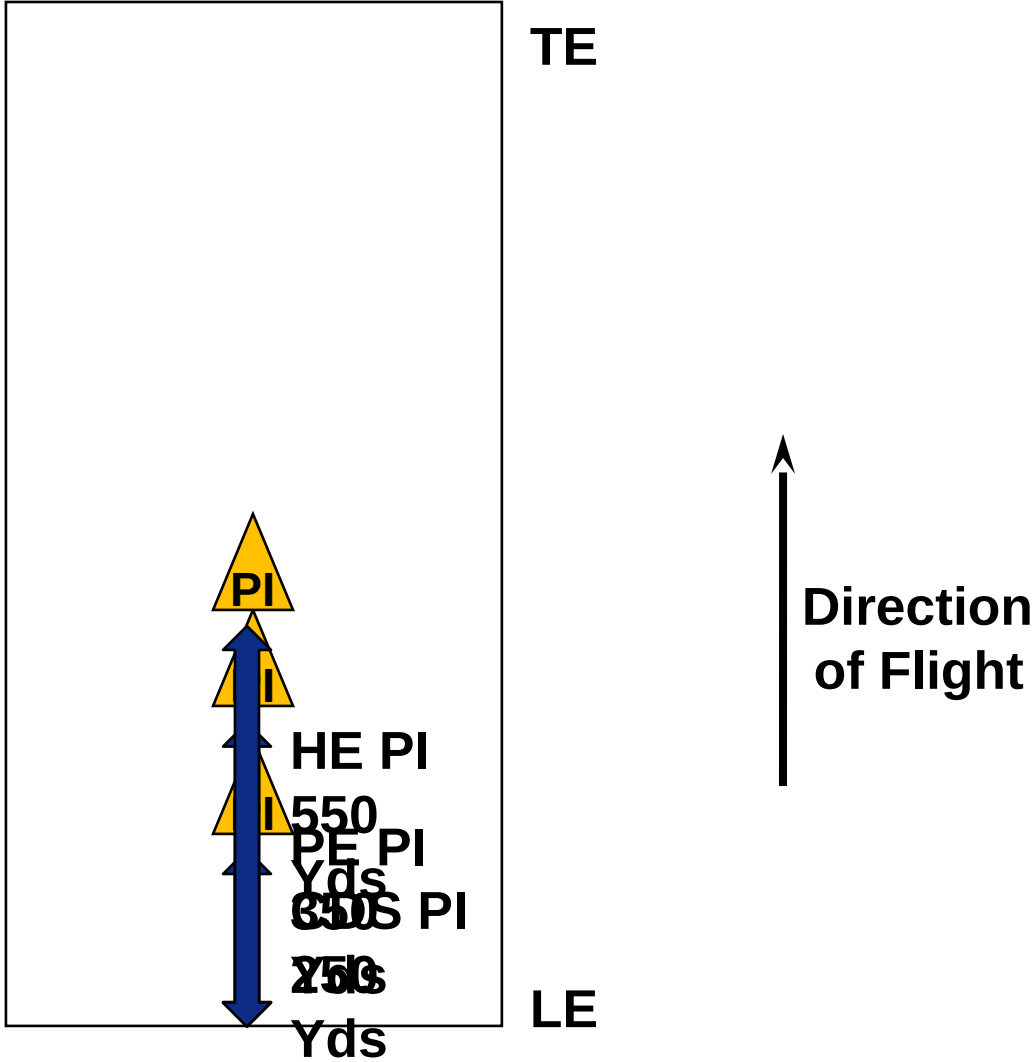
TYPE DROP	DISTANCE FROM APPROACH END	
	DAY	NIGHT
C-130		
CDS	200 yds / 183 m	250 yds / 229 m
PERSONNEL	300 yds / 274 m	350 yds / 320 m
EQUIPMENT	500 yds / 457 m	550 yds / 503 m
C-17	DAY / NIGHT / IMC	NIGHT Pilot Directed Airdrop (PDA)
CDS/ DRAS	225 yds / 206 m	275 yds / 251 m
PERSONNEL	300 yds / 274 m	350 yds / 320 m
EQUIPMENT	500 yds / 457 m	550 yds / 503 m

NOTES:

1. For lateral placement, the PI must be located at least one-half the width of the minimum size DZ (based upon type airdrop and airdrop formation) from the closest side of DZ.
2. The PI may be located anywhere within the surveyed DZ boundaries as long as the minimum required DZ size for that type airdrop and airdrop formation fits within the boundaries, and provide the distance from the leading edge and sides is complied with.
3. JPADS guided systems PI will be the DZ center point unless otherwise coordinated by the supported forces commander as designated supported forces authority by respective Division Commander.



Minimum PI Placement Night Example





Drop Zone

- “MANFAS” method
 - Minimums
 - Altitude
 - Night
 - Formations
 - Additional
 - Safety

KC-130 dropping 12 jumper in 3 passes of 4 from 1500 ft at 1700L

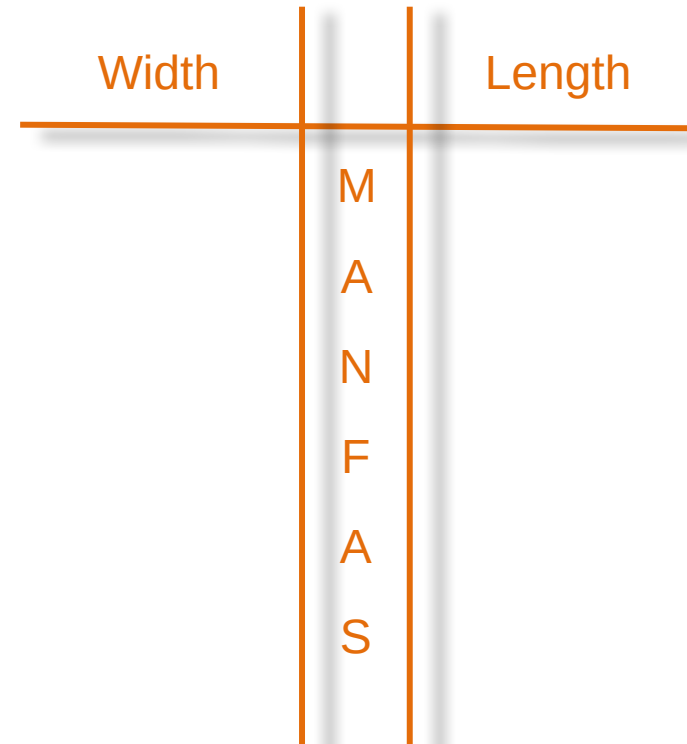
Width		Length
600 yds	M	600 yds
150 yds	A	150 yds
N/A	N	N/A
N/A	F	N/A
N/A	A	225 yds
N/A	S	N/A

Minimum 750 x 975 yds required



Drop Zone

- KC-130 conducting drop of 3 HE platforms 1,700 ft at 2330L





Drop Zone

KC-130 conducting
drop of 3 HE platforms
1,700 ft at 2330L

Width			Length	
600 yds		M	1000 yds	
“each 100 ft above 1100 ft, add 30 yds to the width and 100 yds to width”	→ 180 yds	A	180 yds	
“night single, add 100 yds to width”	→ 100 yds	N	100 yds	←
	N/A	F	N/A	
	N/A	A	800 yds	←
	N/A	S	N/A	

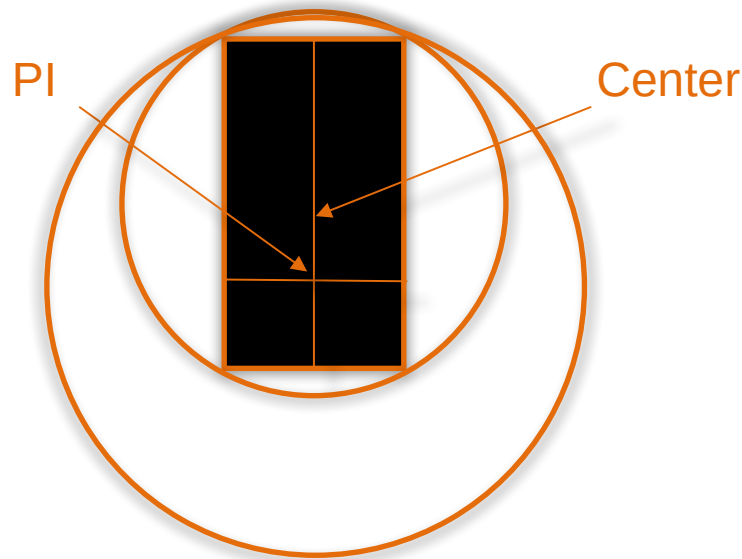
“Sunset to sunrise, add 100 yds to length”
“Add 400 yds to length for each platform >1” 400 x2

Minimum 880 x 2080 yds required



Drop Zone

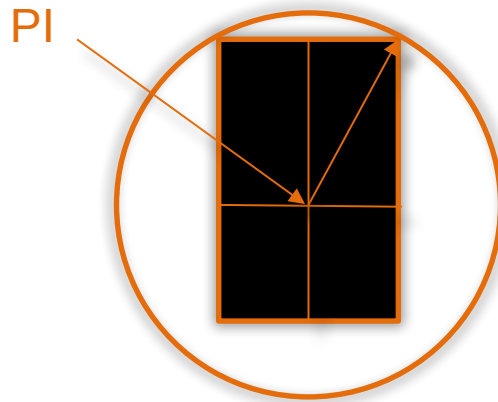
- Circular Drop Zones
 - Center and radius is determined by mission requirements
 - PI normally center for multiple run-in headings



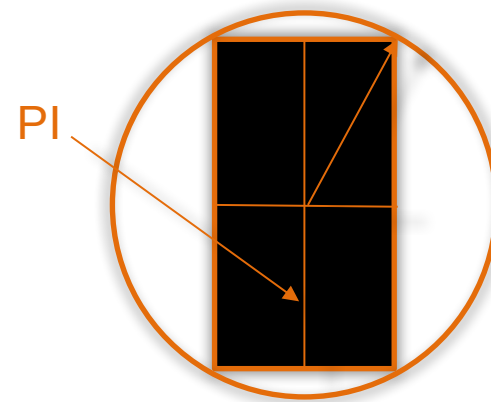


Drop Zone

- Circular Drop Zones



Previously surveyed DZ



Mission specific DZ



Outline

- Drop Zone (DZ)
- Summary



Summary

- Drop Zone (DZ)

Questions / Critiques



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